Exercise 7.4

SOLUTION

Formula ${}^nC_r = \dfrac{n!}{(n-r)!.r!}$

Q.1. Evaluate the following:

i. ${}^{50}C_{50}$

Sol: ${}^{50}C_{50} = \dfrac{50!}{(50-50)!.50!} = \dfrac{1}{0!} = \dfrac{1}{1} = 1$

ii. ${}^{1000}C_0$

Sol: ${}^{1000}C_0 = \dfrac{1000!}{(1000-0)!.0!} = \dfrac{1000!}{1000!.1} = \dfrac{1}{1} = 1$

iii. ${}^{10}C_7$

Sol: ${}^{10}C_7 = \dfrac{10!}{(10-7)!.7!} = \dfrac{10!}{3!.7!} = \dfrac{10.9.8.7!}{3.2.1.7!}$

$= \dfrac{10.9.8}{6} = 120$

iv. ${}^{20}C_{17}$

Sol: ${}^{20}C_{17} = \dfrac{20!}{(20-17)!.17!} = \dfrac{20.19.18.17!}{3!.17!}$

$= \dfrac{20.19.18}{3.2.1} = \dfrac{20.19.\cancel{18}^{3}}{\cancel{6}_{1}} = 1140$

Q.2. (i) If. ${}^{3n}C_2 : {}^nC_2 = 15:1$, Find n

(ii) If ${}^nP_r = 120$ and ${}^nC_r = 20$, Find r

Sol(i): ${}^{3n}C_2 : {}^nC_2 = 15:1$

$\dfrac{{}^{3n}C_2}{{}^nC_2} = \dfrac{15}{1} = {}^{3n}C_2 \div {}^nC_2 = 15$

$\dfrac{3n!}{(3n-2)!.2!} \div \dfrac{n!}{(n-2)!.2!} = 15$

$\dfrac{3n(3n-1)\cancel{(3n-2)!}}{\cancel{(3n-2)!}.2!} \times \dfrac{(n-2)!.2!}{n!} = 15$

$\dfrac{3n(3n-1)}{2!} \times \dfrac{(n-2)!.2!}{n!} = 15$

$3\cancel{n}(3n-1) \times \dfrac{\cancel{(n-2)!}}{\cancel{n}(n-1)\cancel{(n-2)!}} = 15$

$\dfrac{3(3n-1)}{(n-1)} = 15$

$9n-3 = 15(n-1)$

$9n-3 = 15n-15$

$15n-9n = -3+15 = 6n = 12 \Rightarrow n = 2$

(ii) If ${}^nP_r = 120$ and ${}^nC_r = 20$, Find r

Sol: We know that ${}^nC_r \times r! = {}^nP_r$

or $r! = \dfrac{{}^nP_r}{{}^nC_r}$

$r! = \dfrac{120}{20} = 6$

$r! = 3.2.1$

$r! = 3!$

$r = 3$

Q.3. Find the values of n and r, when

i. ${}^nC_r = 56$, ${}^nP_r = 336$

ii. ${}^{n-1}C_{r-1} : {}^nC_r : {}^nC_r = 1:3:7$

i. ${}^nC_r = 56$, ${}^nP_r = 336$

Sol: ${}^nC_r = 56$, ${}^nP_r = 336$

$\dfrac{n!}{(n-r)!.r!} = 56$ _____(I)

$\dfrac{n!}{(n-r)!} = 336$ _____(II)

Divide (II) by (I)

$\dfrac{n!}{(n-r)!} \div \dfrac{n!}{(n-r)!.r!} = \dfrac{336}{56}$

$\dfrac{\cancel{n!}}{\cancel{(n-r)!}} \times \dfrac{\cancel{(n-r)!}.r!}{\cancel{n!}} = 6$

$r! = 6 \Rightarrow r! = 3! \Rightarrow \boxed{r = 3}$

Put $r = 3$ in (II)

$\dfrac{n!}{(n-3)!} = 336$

$\dfrac{n(n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}} = \dfrac{336}{56}$

$n(n-1)(n-2) = 8.7.6 \Rightarrow n = 8$

$r = 3$ and $n = 8$

ii. ${}^{n-1}C_{r-1} : {}^nC_r : {}^{n+1}C_{r+1} = 1:3:7$

Sol: ${}^{n-1}C_{r-1} : {}^nC_r : {}^{n+1}C_{r+1} = 1:3:7$

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$^{n-1}C_{r-1} : {}^nC_r = 1:3$

$^{n-1}C_{r-1} \div {}^nC_r = 1 \div 3$

$$\frac{(n-1)!}{(n-1-r+1)!(r-1)!} \div \frac{n!}{(n-r)!r!} = \frac{1}{3}$$

$$\frac{(n-1)!}{(n-r)!(r-1)!} \times \frac{(n-r)!r!}{n!} = \frac{1}{3}$$

$$\frac{(n-1)!}{(r-1)!} \times \frac{r(r-1)}{n(n-1)!} = \frac{1}{3}$$

$$\frac{r}{n} = \frac{1}{3} \Rightarrow 3r = n \text{ ------I}$$

Now, $^nC_r : {}^{n+1}C_{r+1} = 3:7$

$^nC_r \div {}^{n+1}C_{r+1} = 3 \div 7$

$$\frac{n!}{r!(n-r)!} \div \frac{(n+1)!}{(r+1)!(n+1-r-1)!} = 3 \div 7$$

$$\frac{n!}{n!(n-r)!} \times \frac{(r+1).r! \times (n-r)!}{(n+1).n!} = \frac{3}{7}$$

$$\frac{r+1}{n+1} = \frac{3}{7} \Rightarrow 7(r+1) = 3(n+1) \text{ ----II}$$

Put I in II

$$7(r+1) = 3(3r+1) \Rightarrow 7r+7 = 9r+3$$

$$9r-7r = 7-3 \Rightarrow 2r = 4 \Rightarrow \boxed{r=2}$$

Put in I $\Rightarrow n = 3(2) \Rightarrow \boxed{n=6}$

Q.4. Prove that

i. $^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

Sol: $^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

L.H.S $= {}^nC_r + {}^nC_{r-1}$

$$= \frac{n!}{(n-r)!r!} + \frac{n!}{(n-r+1)!(r-1)!}$$

$$= \frac{n!}{(n-r)!r(r-1)!} + \frac{n!}{(n-r+1)(n-r)!(r-1)!}$$

$$= \frac{n!}{(n-r)!(r-1)!}\left[\frac{1}{r} + \frac{1}{n-r+1}\right]$$

$$= \frac{n!}{(n-r)!(r-1)!}\left[\frac{n-r+1+r}{r(n-r+1)}\right]$$

$$= \frac{(n+1)n!}{(n-r+1)(n-r)!r(r-1)!} = \frac{(n+1)!}{(n-r+1)!r!}$$

$$= {}^{n+1}C_r = RHS$$

ii. $r.{}^nC_r = (n-r+1)\,{}^nC_{r-1}$

Sol: $r.{}^nC_r = (n-r+1)\,{}^nC_{r-1}$

L.H.S $= r.{}^nC_r = r.\dfrac{n!}{(n-r)!r!}$ ——— (I)

R.H.S $= (n-r+1).{}^nC_{r-1}$

$$= (n-r+1).\frac{n!}{(n-r+1)!.(r-1)!}$$

$$= \left((n-r+1)\times\frac{(n-r)!}{(n-r)!}\right).\left(\frac{n!}{(n-r+1)!.(r-1)!}\times\frac{r}{r}\right)$$

$$= \frac{(n-r+1)!}{(n-r)!} \times \frac{r.n!}{(n-r+1)!r(r-1)!} = r.\frac{n!}{(n-r)!r!}$$

$= r.{}^nC_r$ = L.H.S

Q.5. Prove that product of r consecutive integers is divisible by $r!$.

Sol: Proof:

Let the r consecutive integers be

$(n+1),(n+2),(n+3)...(n+r)$ then, their product

$$= (n+1)(n+2)(n+3).....(n+r)$$

$$= \frac{n!(n+1)(n+2)(n+3).....(n+r)}{n!} \text{ ('×' \& '÷' by } n!)$$

$$= \frac{(n+r).....(n+3)(n+2)(n+1)n!}{n!}$$

$$= \frac{(n+r)!}{n!} = \frac{(n+r)!}{n!} \times \frac{r!}{r!} = r!.\frac{(n+r)!}{r!(n+r-r)!}$$

$= r! \times {}^{n+r}C_r$ which is Divisible by $r!$.

Since $^{n+r}C_r$ is positive integer.

Q.6. In how many ways can five subjects be selected out of eight subjects to select a course program?

Sol: Total subjects: $= n = 8$

Subjects to be selected: $= r = 5 \Rightarrow r = 5$

Total ways: $= {}^nC_r = {}^8C_5$

$$= \frac{8!}{(8-5)!.5!} = \frac{8.7.6.5!}{3!.5!} = \frac{8.7.6}{6}$$

$$= 56$$

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Q.7. **Find the number of possible arrangements of vowel letters from the English alphabet?**

Sol: Vowels are a, e, i, o, u

$\Rightarrow \quad n = 5, \quad r = 5$

Total ways ${}^5P_5 = 5!$

$= 5.4.3.2.1 = 120$

Q.8. **In how many ways 3 dishes of Desi food and 2 dishes of Chinese food be selected from 6 dishes of Desi food and 8 dishes of Chinese food?**

Sol:

Total dishes of Desi food $= 6 \Rightarrow n = 6$

To be selected $= 3 \Rightarrow r = 3$

Total dishes of Chinese food $= 8 \Rightarrow n = 8$

To be selected $= 2 \Rightarrow r = 2$

Total ways $= {}^6C_3 . {}^8C_2$

$$= \frac{6!}{(6-3)!3!} \cdot \frac{8!}{(8-2)!.2!} = \frac{6.5.4.\cancel{3!}}{3!.\cancel{3!}} \cdot \frac{8.7.\cancel{6!}}{\cancel{6!}.2!}$$

$$= \frac{6.5.4}{6} \times \frac{8.7}{2} = 20 \times 28 = 560$$

Q.9. **From a standard deck of 52 playing cards, there are 26 black cards and 26 red cards. How many different ways can eight cards be selected if 3 are black and the remaining 5 are red?**

Sol:

Total black cards $= 26$

Total red cards $= 26$

To be arranged cards $= 8$

To be arranged black cards $= 3$

To be arranged red cards $= 8 - 3 = 5$

Total possible ways $= {}^{26}C_3 \cdot {}^{26}C_5$

$$= \frac{26!}{(26-3)!.3!} \cdot \frac{26!}{(26-5)!.5!}$$

$$= \frac{26.25.\cancel{24}^{4}.\cancel{23!}}{\cancel{23!}.\cancel{6}} \times \frac{26.25.24.23.22.\cancel{21!}}{\cancel{21!}.120}$$

$$= 2600 \times 5780$$

$$= 171,028,000$$

Q.10. **A bag contains 8 red balls, 7 green balls. Find the total number of possible ways in which five balls are selected in a way:**

Total Red balls $= 8$

Total Green balls $= 7$

Balls to be selected $= 5$

i. **3 red and 2 green**

Sol: Total ways $= {}^8C_3 . {}^7C_2$

$$= \frac{8!}{(8-3)!.3!} \times \frac{7!}{(7-2)!.2!} = \frac{8!}{5!3!} \cdot \frac{7!}{5!.2!}$$

$$= \frac{8.7.6.\cancel{5!}}{\cancel{5!}.3.2.1} \cdot \frac{7.6.\cancel{5!}}{\cancel{5!}.2.1} = 56.21 = 1176$$

ii. **1 red and 4 green**

Sol: Total ways $= {}^8C_1 . {}^7C_4$

$$= \frac{8!}{(8-1)!.1!} \times \frac{7!}{(7-4)!.4!}$$

$$= \frac{8.\cancel{7!}}{\cancel{7!}.1!} \times \frac{7.6.5.\cancel{4!}}{3!.\cancel{4!}} = 8 \times \frac{7.\cancel{6}^{3}.5}{\cancel{6}} = 8 \times 7 \times 5 = 280$$

iii. **4 red and 1 green**

Sol: Total ways $= {}^8C_4 . {}^7C_1$

$$= \frac{8!}{(8-4)!.4!} \times \frac{7!}{(7-1)!.1!}$$

$$= \frac{8.7.6.5.\cancel{4!}}{4.!\cancel{4!}} \times \frac{7.\cancel{6!}}{\cancel{6!}.1} = \frac{8.7.6.5}{4.3.2.1} \times 7 = 490$$

iv. **All the red balls**

Sol: Total ways $= {}^8C_5$

$$= \frac{8!}{(8-5)!.5!} = \frac{8.7.6.\cancel{5!}}{3!.\cancel{5}!} = \frac{8.7.\cancel{6}}{\cancel{6}} = 56$$

Q.11. **How many diagonals and triangles can be formed by joining the vertices of the polygon having 15 sides**

Sol: For diagonal $= {}^nC_2 - n$

For triangle $= {}^nC_3$

Total Side of polygon = 15 $\Rightarrow n = 15$

Numbers of Diagonals $= {}^{15}C_2 - 15$

$$= \frac{15!}{(15-2)!.2!} - 15 = \frac{15.\cancel{14}^{7}.\cancel{13!}}{\cancel{13!}.\cancel{2}.1} - 15$$

$$= 105 - 15 = 90$$

Numbers of Triangles $= {}^nC_3 = {}^{15}C_3$

$$= \frac{15!}{(15-3)!.3!} = \frac{15.14.13.\cancel{12!}}{\cancel{12!}.3.2.1} = 455$$

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Q.12. **Find the number of sides of a polygon if the number of its diagonals is 104**

Sol: Let Number of sides of polygon $= n$

Given Diagonals $= 104$

Number of Diagonals $= {}^{n}C_{2} - n$

$$104 = {}^{n}C_{2} - n$$

$$104 = \frac{n!}{(n-2)!.2!} - n = \frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}.2!} - n$$

$$104 = \frac{n(n-1)}{2} - n$$

$$104 = n\left[\frac{n-1}{2} - 1\right]$$

$$104 = n\left[\frac{n-1-2}{2}\right]$$

$$104 = \frac{n}{2}[n-3]$$

$$2 \times 104 = \cancel{2} \times \frac{n}{\cancel{2}}[n-3]$$

$$208 = n^2 - 3n$$

$$n^2 - 3n - 208 = 0$$

$$n^2 - 16n + 13n - 208 = 0$$

$$n(n-16) + 13(n-16) = 0$$

$$(n-16)(n+13) = 0$$

$n - 16 = 0$, $n + 13 = 0$

$\boxed{n = 16}$, $n = -13$ Impossible

Q.13 **How many triangles can be formed by joining 15 points, 6 of which lie on the same straight line?**

Sol: Total points $= 15 \Rightarrow n = 15$

Total triangles formed by

15 points $= {}^{n}C_{3} = {}^{15}C_{3}$

$$\frac{15!}{(15-3)!.3!} = \frac{15.14.13.\cancel{12!}}{\cancel{12!}.3!} = \frac{15.14.13}{6} = 455$$

Since 6 points lie on the straight line. Choosing any '3' of these

6 points will not form a triangle. The number of such combinations $= {}^{6}C_{3}$

$$\frac{6!}{(6-3)!.3!} = \frac{6.5.4.\cancel{3!}}{3!.\cancel{3!}} = \frac{\cancel{6}.5.4}{\cancel{6}} = 20$$

Total triangles $= 455 - 20 = 435$

Q.14. **The members of a club are 10 boys and 8 girls. In how many ways can a committee of 6 boys and 3 girls be formed?**

Sol. Boys: $n = 10$, $r = 6$

Girls: $n = 8$, $r = 3$

Number of ways of committee $= {}^{10}C_{6} \times {}^{8}C_{3}$

$$= \frac{10!}{(10-6)!.6!} \times \frac{8!}{(8-3)!.3!}$$

$$= \frac{10!}{4!.6!} \times \frac{8!}{5!.3!}$$

$$= \frac{10.9.8.7.\cancel{6!}}{4.3.2.1.\cancel{6!}} \times \frac{8.7.6.\cancel{5!}}{\cancel{5!}.3.2.1} = \frac{10.9.8.7}{4.3.2.1} \times \frac{8.7.6}{3.2.1}$$

$$= 210 \times 56 = 11760$$

Q.15. **How many committees of 7 members can be chosen from a group of 10 persons when each committee must include 2 particular persons?**

Sol. $n = 10$; $r = 7$

For two particular persons

$\Rightarrow$ $n = 8$; $r = 5$

Total committees $= {}^{n}C_{r} = {}^{8}C_{5}$

$$= \frac{8!}{(8-5)!.5!} = \frac{8.7.6.\cancel{5!}}{3!.\cancel{5!}} = \frac{8.7.6}{3.2.1} = 56$$

Q.16. **In how many ways can a cricket team of 11 players be selected out of 17 players? How many of them will include a particular player?**

Sol. Total Players $= 17$

To be selected $= 11$

$n = 17$; $r = 11$

Total ways $= {}^{17}C_{11}$

$$= \frac{17!}{(17-11)!.11!} = \frac{17.16.15.14.13.12.\cancel{11!}}{6.!.\cancel{11!}}$$

$$= \frac{8,910,720}{720}$$

$= 12,376$

When a particular player is include

$\Rightarrow$ $n = 16 = 17 - 1$; $r = 10 = 11 - 1$

Total ways $= {}^{16}C_{10}$

$$= \frac{16!}{(16-10)!.10!} = \frac{16.15.14.13.12.11.\cancel{10!}}{6!.\cancel{10!}}$$

$$= \frac{5765,760}{720} = 8008$$

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Q.17. There are 6 men and 8 women members of a club. How many committees of seven can be formed?

Sol. Men = 6 , Women = 8

Total members = 7

i. **With 3 women**

Total committees $= {}^8C_3 \times {}^6C_4$

$$= \frac{8!}{(8-3)!.3!} \times \frac{6!}{(6-4)!.4!} = \frac{8.7.6.\cancel{5!}}{\cancel{5!}.3.2.1} \times \frac{6.5.\cancel{4!}}{2!.\cancel{4!}} = \frac{8.7.\cancel{6}}{\cancel{6}} \times \frac{\cancel{6}.5}{\cancel{2}} = 56 \times 15 = 840$$

ii. **with at most 3 women**

Sol. Total committees $= \left({}^8C_1 \times {}^6C_6\right) + \left({}^8C_2 \times {}^6C_5\right) + \left({}^8C_3 \times {}^6C_4\right)$

$$= \frac{8!}{(8-1)!.1!} \times \frac{6!}{(6-6)!.6!} + \frac{8!}{(8-2)!.2!} \times \frac{6!}{(6-5)!.5!} + \frac{8!}{(8-3)!.3!} \times \frac{6!}{(6-4)!.4!}$$

$$= \frac{8.\cancel{7!}}{\cancel{7!}.1!} \times \frac{\cancel{6!}}{(6-6)!.\cancel{6!}} + \frac{8.7.\cancel{6!}}{\cancel{6!}.2!} \times \frac{6.\cancel{5!}}{1!.\cancel{5!}} + \frac{8.7.6.\cancel{5!}}{\cancel{5!}.3!} \times \frac{6.5.\cancel{4!}}{2!.\cancel{4!}}$$

$$= \left(8 \times \frac{1}{0!}\right) + \left(\frac{8.7}{2} \times 6\right) + \left(\frac{8.7.6}{3.2.1} \times \frac{6.5}{2.1}\right) = 8 + 168 + 840 = 1016$$

iii. **with at least 5 women**

Sol. Total ways $= {}^8C_5 \times {}^6C_2 + {}^8C_6 \times {}^6C_1 + {}^8C_7 \times {}^6C_0$

$$= \left(\frac{8!}{(8-5)!.5!} \times \frac{6!}{(6-2)!.2!}\right) + \left(\frac{8!}{(8-6)!6!} \times \frac{6!}{(6-1)!.1!}\right) + \left(\frac{8!}{(8-7)!.7!} \times \frac{6!}{(6-0)!.0!}\right)$$

$$= \left(\frac{8.7.6.\cancel{5!}}{3!.\cancel{5!}} \times \frac{6.5.\cancel{4!}}{\cancel{4!}.2!}\right) + \left(\frac{8.7.\cancel{6!}}{2!.\cancel{6!}} \times \frac{6.\cancel{5!}}{\cancel{5!}.1!}\right) + \left(\frac{8.\cancel{7!}}{1!.\cancel{7!}} \times \frac{\cancel{6!}}{\cancel{6!}.0!}\right)$$

$$= \left(\frac{8.7.6}{6} \times \frac{6.5}{2}\right) + \left(\frac{8.7}{2} \times 6\right) + (8 \times 1) = 840 + 168 + 8 = 1016$$

Q.18. There are three section in a question paper, each section has 3 questions. A student has to solve all 5 questions, choosing at least one question from each section. In how many ways can the student make his choice?

Sol: Total section $= 3 \Rightarrow$ Total Options $= 3! = 6$

Each section has questions $= 3$

Total number of ways:

(Option – I)

$${}^3C_1 \times {}^3C_1 \times {}^3C_3 = \frac{3!}{(3-1)!.1!} \times \frac{3!}{(3-1)!.1!} \times \frac{3!}{(3-3)!.3!} = \frac{3.\cancel{2!}}{\cancel{2!}.1!} \times \frac{3.\cancel{2!}}{\cancel{2!}.1!} \times \frac{\cancel{3!}}{0!.\cancel{3!}} = 3 \times 3 \times 1 = 9$$

(Option – II)

$${}^3C_1 \times {}^3C_2 \times {}^3C_2 = \frac{3!}{(3-1)!.1!} \times \frac{3!}{(3-2)!.2!} \times \frac{3!}{(3-2)!.2!} = \frac{3.\cancel{2!}}{\cancel{2!}.} \times \frac{3.\cancel{2!}}{1!.\cancel{2!}} \times \frac{3.\cancel{2!}}{1!.\cancel{2!}} = = 3 \times 3 \times 3 = 27$$

(Option – III)

$${}^3C_1 \times {}^3C_3 \times {}^3C_1 = \frac{3!}{(3-1)!.1!} \times \frac{3!}{(3-3)!.3!} \times \frac{3!}{(3-1)!.1!} = 3 \times 1 \times 3 = 9$$

(Option – IV)

$${}^3C_2 \times {}^3C_1 \times {}^3C_2 = \frac{3!}{(3-2)!.2!} \times \frac{3!}{(3-1)!.1!} \times \frac{3!}{(3-2)!.2!} = \frac{3.\cancel{2!}}{1!.\cancel{2!}} \times \frac{3.\cancel{2!}}{\cancel{2!}.1!} \times \frac{3.\cancel{2!}}{1!.\cancel{2!}} = = 3 \times 3 \times 3 = 27$$

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(Option – V)

$${}^3C_3 \times {}^3C_1 \times {}^3C_1 = \frac{3!}{(3-3)!.3!} \times \frac{3!}{(3-1)!.1!} \times \frac{3!}{(3-1)!.1!} = \frac{\cancel{3!}}{0!.\cancel{3!}} \times \frac{3.\cancel{2!}}{\cancel{2!}.1!} \times \frac{3.\cancel{2!}}{\cancel{2!}.1!} = 1 \times 3 \times 3 = 9$$

Total Ways: $9 + 27 + 9 + 27 + 9 = 108$

Q.19. Consider a cryptographic system that generates an 8 characters password. Each character in the password can be either a lowercase letter (a – f) or a digit (0 – 5). How many passwords can be generated if each password must contain exactly 5 lowercase letters and 3 digits?

a. with repetition allowed

Sol. Lower case letters $= a - f$

Total options $= 6$

Digits $= (0-5)$

Total options $= 6$

Letters to be selected $= 5$

Digits to be selected $= 3$

Total Characters in Password $= 8$

Options for letters $= {}^8C_5$

$$= \frac{8!}{(8-5)!.5!} = \frac{8.7.6.\cancel{5!}}{3!.\cancel{5!}} = \frac{8.7.\cancel{6}}{\cancel{6}} = 56$$

Number of Password $= {}^8C_5 = 56 \times 6^3 \times 6^3 = 56 \times 7776 \times 216$

$= 56 \times 1679616 = 94058496$

b. Without repetition

Sol: $56 \times {}^6P_5 \times {}^6P_3$

$$= 56 \times \frac{6!}{(6-5)!} \times \frac{6!}{(6-3)!} = 56 \times \frac{6.5.4.3.2.1}{1!} \times \frac{6.5.4.\cancel{3!}}{\cancel{3!}} = 56 \times 72 \times 120 = 4838400$$

Q.20. On Defense Day, Teacher I compiles a list of 10 distinct national songs, while Teacher II prepares a separate list of 10 different national songs (with no overlap between the two lists). The principal needs to select 3 songs from Teacher I's list, and 3 songs from Teacher II's list.

Determine the number of possible selection methods when:

I. The order/sequence of the selected songs is important.

Sol: Teachers I one prepares songs $= 10$

To be selected by principle $= 3$

Teacher II prepares songs $= 10$

To be selected by principle $= 3$

Total ways $= {}^{10}P_3 \cdot {}^{10}P_3$

$$= \frac{10!}{(10-3)!}.\frac{10!}{(10-3)!} = \frac{10.9.8.\cancel{7!}}{\cancel{7!}} \times \frac{10.9.8.\cancel{7!}}{\cancel{7!}} = 10 \times 9 \times 8 \times 10 \times 9 \times 8 = 518,400$$

II. The order/sequence of the selected songs is not important.

Sol. Teachers I one prepares songs $= 10$

To be selected by principle $= 3$

Teacher II prepares songs $= 10$

To be selected by principle $= 3$

Total ways when the sequence of the selected songs does not matter

$$= {}^{10}C_3 \cdot {}^{10}C_3 = \frac{10!}{(10-3)!.3!}.\frac{10!}{(10-3)!.3!} = \frac{10.\cancel{9}^3.\cancel{8}^4.\cancel{7!}}{\cancel{7!}.\cancel{3}.\cancel{2}.1} \times \frac{10.\cancel{9}^3.\cancel{8}^4.\cancel{7!}}{\cancel{7!}.\cancel{3}.\cancel{2}.1} = 120 \times 120 = 14400$$

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